Myles Lozano

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Aerospace and mechanical engineering student with hands-on experience supporting military engine hardware and developing field-ready systems. Bringing a strong work ethic, rapid learning mindset, and eagerness to take on challenging technical problems.

Professional Experience

TriPoint Innovations, LLC — **Stillwater, OK**

❖ **Co-Founder & Product Lead** — **November 2025 – Present**

- Built 3 prototype versions and 4 ESP32-based LoRa/GPS tracking nodes for rugged offline field communication, waypoint navigation, boundary alerts, and SOS signaling.
- Field-tested Hunter Radar across 2 outdoor trials, achieving 3.5-mile LoRa range and 12+ hour battery life.
- Iterated 3 enclosure designs to improve antenna placement, GPS lock, battery packaging, durability, and field usability.
- Developed architecture, requirements, and validation plan for a low-power GPS/LoRa situational awareness device.
- Secured a $75,000 USDA Grant for prototype-to-production manufacturing, covering injection molding, electronics testing, tooling, and assembly equipment.

OSU New Product Development Laboratory — **Stillwater, OK**

❖ **Mechanical Engineering Intern** — **August 2025 – Present**

- Designed CAD models in SolidWorks, prototypes, and test fixtures across 12 client hardware projects.
- Iterated 8+ CAD models through client feedback, DFM review, and prototype test plans and analysis.
- Conducted 3 CFD studies to evaluate airflow, fluid behavior, and design feasibility for a confidential reverse-engineering project.
- Built 7 test fixtures and created patent drawings for hardware concepts moving toward IP filing.

GE Aerospace – Strother — **Winfield, KS**

❖ **Manufacturing Engineering Intern (Military MSO)** — **May 2026 – August 2026**

- Created and tested 18+ VGT inspection tooling concepts in Autodesk Inventor for sub-thousandth turbine inspection.
- Reduced inspection time by developing tooling for dimensional checks on features with tolerances up to 0.0006 in.
- Led Stage 1 nozzle flow investigation linking TBC thickness, EDM cutback, geometry variation, and flow performance.
- Identified 3 hardware failure modes and created 5 process improvements across engineering, quality, production, component, and maintenance teams.

Springfield Remanufacturing Corp. — **Springfield, MO**

❖ **Product Engineering Intern** — **May 2025 – August 2025**

- Led teardown and inspection of one 6.7L Cummins engine and four additional engines for remanufacturing feasibility.
- Validated aftermarket components against OEM standards, documenting failure modes and recommendations with potential to reduce per-unit costs by 30%.

Related Projects

Oklahoma State University CEAT — **Stillwater, OK**

❖ **Engineering Project Lead** — **Fall 2024 – Present**

- Led 8-person engineering team and 30+ laborers building a 60 ft x 18 ft x 12 ft automated mechanical display.
- Integrated 8+ motors and actuators across 6 moving linkages with relay logic, timing controls, and enclosure wiring.
- Used CAD, motion studies, and FEA to identify failure causes in 3 metal components, preventing $4,500 in fabrication rework.
- Placed 1st out of 12 contestants in front of over 200,000 people at OSU's Homecoming Celebration.

Technical & Software Skills

- **Design (CAD):** SolidWorks, Inventor, Fusion, 3D slicers, GD&T, DFM
- **Analysis:** MATLAB, Oscilloscopes, RF Analysis, CAD assemblies, SolidWorks Flow, engineering drawings, CFD, FEA
- **Mechanical Hardware:** enclosure design, tooling, test fixtures, injection molding design, prototype manufacturing
- **Electrical/Controls:** circuit fundamentals, sensors, motors, actuators, relay logic, wiring, embedded systems
- **Communication Systems:** ESP32, LoRa, GPS, BLE, Radar, low-power hardware integration, mesh tracking systems
- **Tools**: Python, GitHub, Excel, KiCad, FDM printing

Education

- **Aerospace Engineering; Mechanical Engineering** — **Fall 2024 – Spring 2028 (Expected)**
  Minor: Electrical Engineering
  Oklahoma State University, Stillwater, OK
  GPA: 3.70

Honors & Awards

- *OSU Top 20 Freshman* — **September 2025**
- *Impact Award, GE Aerospace* — **July 2026**
- *CEAT Scholar* — **August 2024–Present**
- *Third Place Overall, Hack OSU Hackathon* — **November 2025**
- *Dean's Honor Roll* — **Fall 2024–Spring 2026**
- *CORD Gold Certificate, 300+ Community Service Hours* — **January 2026**
- *Citizenship Award, Republic, Missouri* — **May 2024**
- *Emerging Leader, FarmHouse Fraternity* — **February 2025**

Leadership & Volunteer Experience

- *External Vice President, FarmHouse Fraternity* — **January 2026–Present**
- *International Undergraduate Growth Committee* — **June 2026–Present**
- *Activate, Oklahoma Farm Bureau Business Accelerator* — **May 2026–Present**
- *Riata Startup Experience Pre-Accelerator* — **November 2025–Present**
- *Zink Center for Competitive Innovation* — **November 2025–Present**
- *Monster Energy College Ambassador Team* — **August 2025–Present**
- *Peer Minister, Zion Lutheran Church* — **January 2026–Present**
- *OSU Student Council, Freshman Representative Council* — **August 2024–May 2025**
- *CEAT Student Council, CEAT Freshman Council* — **August 2024–May 2025**
- *Into the Streets Volunteer and Site Leader* — **Fall and Spring 2024–2026**
- *FarmHouse Leadership Institute* — **January 2026**
- *To Be and Become Leadership Retreat* — **May 2025**
- *ASME Student Member* — **August 2024–Present**
- *AIAA Student Member* — **August 2024–Present**
- *Oklahoma Secondary School Activities Association Official* — **May 2024–Present**
- *Missouri State High School Activities Association Official* — **May 2022–Present**